Problem 8.1 (MATLAB)
Non-parametric spectral estimation

In this problem is studied the classical methods for non-parametric ("black-box") spectral estimation

- The Periodogram estimate with settings: signal length
- The Bartlett estimate with settings: signal length and number of sub-blocks
- The Welch estimate with settings: signal length, number of sub-blocks, window function and optionally block overlap percentage
- The Blackman-Tukey estimate with settings: signal length, window width and window function

As for implementation of the methods, you can

- Implement the methods by programming m-file functions
- Use the m-file functions made available on Blackboard
- Use MATLAB built-in functions where applicable

Define various generic signals, e.g.

- Pure white noise signal
- Band-limited white noise signal
- Pure harmonic signal (one or several sinusoids)
- Two or more close sinusoids
- Combinations of the above
- Other signals as desired

Spectral estimation:

- Estimate PSD's for the defined signals using the 4 non-parametric methods with the chosen implementations
- Vary the settings ("parameters") for the methods
- Compare the methods
- Study concepts such as bias, variance and frequency resolution where relevant
- Compare with theoretical properties for the methods